

Problem-Solution Fit

Date	14 July 2026
Team ID	
Project Name	Automated Credit Card Application Screening using ML
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] - Retail Banks & Financial Institutions - Compliance Officers & Risk Analysts - Credit Card Applicants (End-users) - Fintech Startups	6. CUSTOMER LIMITATIONS [CL] - Limited credit history for new applicants - Inability to process high volume manually - Subjective bias in human judgment - Privacy concerns over financial data	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Review: High accuracy but extremely slow and expensive. Scorecards: Reliable but rigid; fails to capture non-linear patterns. Global ML Models: Fast, but often lack local explainability (SHAP).
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Long waiting times for customers (Daily) - High operational cost of manual screening - Missed revenue due to processing delays - Inconsistent labeling of payment risks	9. PROBLEM ROOT / CAUSE [RC] - Fragmented applicant data sources - Manual normalization of income/employment - Complexity of multi-class payment statuses - Lack of real-time screening infrastructure	7. BEHAVIOR + ITS INTENSITY [BE] - Customers expect 24/7 instant decisions - High churn rate if approval takes > 48hrs - Compliance needs "Audit Trails" for rejections
3. TRIGGERS TO ACT [TR] - Digital banking transformation - Rise in credit card application volume - Need for error-free risk assessment - Competitive pressure from Neo-banks	10. YOUR SOLUTION [SL] An automated screening app using ML (XGBoost, Random Forest). Features Grid Search optimization for accuracy, SHAP values for explainability, and a Flask dashboard for real-time visualization of applicant risks and approval trends.	8. CHANNELS OF BEHAVIOR [CH] Online: Mobile banking, Flask Web Portal, IBM Watson Cloud. Offline: Bank branch tablets, Customer service centers.
4. EMOTIONS BEFORE / AFTER [EM] Before: Anxiety over approval, frustration with paperwork, uncertainty of decision factors. After: Instant clarity, confidence in fair assessment (Explainability), streamlined user experience.		